

~~diff~~

Difference equation:

$$x_{\text{new}} = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_{\text{new}} = x_0 - \frac{(e^{x_0} - 2)}{e^{x_0}}$$

$$= 1 - \frac{(e - 2)}{e}$$

$$x_{\text{new}} = \frac{2}{e}$$